

2(a)	$65 = \rho g V$ or $65 = mg$ and $m = \rho V$	C1
	$\rho = 65 / (9.81 \times 7.5)$ $= 0.88 \text{ kg m}^{-3}$	A1
2(b)(i)	acceleration = 9.8 m s^{-2}	A1
2(b)(ii)	air resistance (acting on sphere) increases	B1
	resultant force (on sphere) decreases	B1
	(magnitude of) acceleration decreases	B1
2(c)	$F = ma$ $= 4.0 \times 1.9$ $(= 7.6 \text{ N})$	C1
	resistive force = $(4.0 \times 9.81) - (4.0 \times 1.9)$ $= 32 \text{ N}$	A1